

## DATA CLEANING AND PROCESSING

1. To select required columns

```
df_selected = df[["col1", "col2", "col3"]]
```

2. To drop unwanted columns

```
df_selected = df.drop(columns = ["col1"])
```

3. To filter rows for given conditions

**# single condition**

```
df_filtered = df[df["Age"] > 25]
```

**# Multiple conditions**

```
df_filtered = df[(df["Age"] > 25) & (df["Department"] == "Sales")]
```

**# Remove rows where condition**

```
df_filtered = df[df["Salary"].notna()]
```

4. Rename Columns

```
df_renamed = df.rename(columns={  
    "Name"      : "Employee_Name",  
    "Salary"    : "Annual_Salary",  
    "Department" : "Dept"  
})
```

5. To add new calculated column

```
df["Monthly_Salary"] = df["Salary"] / 12  
df["Tax"]             = df["Salary"] * 0.2  
df["Net_Salary"]      = df["Salary"] - df["Tax"]
```

6. To sort data

```
# Ascending  
df_sorted = df.sort_values("Salary")  
  
# Descending  
df_sorted = df.sort_values("Salary", ascending=False)  
  
# Multiple columns  
df_sorted = df.sort_values(["Department", "Salary"],  
                           ascending=[True, False])
```

## 7. Handling missing values

```
# Count missing values
print(df.isnull().sum())

# Fill missing Salary with mean
df["Salary"] = df["Salary"].fillna(df["Salary"].mean())

# Fill missing text with a label
df["Department"] = df["Department"].fillna("Unknown")

# Drop rows with any missing values
df_clean = df.dropna()

# Drop rows where specific column is null
df_clean = df.dropna(subset=["Salary"])
```

## 8. Remove duplicates

```
# Remove completely duplicate rows
df_unique = df.drop_duplicates()
```

## 9. To count no of values in a column

```
df.col.value_counts()
```

## 10. To find unique values

```
df[col].unique()
```

## 11. To create new dataframe

```
# Create new dataframe with only females and selected columns
df_females = df[df['Gender'] == 'Female'][['Salary', 'Age', 'Department']]
```

### Ques 2:

Load the employee\_salary dataset and perform ETL operations on the given data.

### What you can do:

Find all columns, data types, shape, first 10 rows

- Generate the complete statistical summary and answer the following
  - a) The maximum age
  - b) The lowest salary
  - c) The department with maximum employees
- Filter employees by age > 30
- Calculate annual salary

- Create a new data frame with only females, their salary, age and department
- Find the unique values of column education level
- Find the count of people belonging to each city
- Filter the male employees with salary more than 50000